

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2583–2595; original printed pp. 86–87; physical source-PDF p. 77; recomposed running p. 69. This bounded unit has been independently source-reviewed against the corrected French TeX and direct scan; cumulative integration remains pending. Line 2596 is blank; the construction of the double complex begins at line 2597 and is excluded.

The proof is left to the reader.¹

Let us recall a notation introduced in Exposé V.

Notation. — Let $X^\bullet$ and $Y^\bullet$ be two complexes. We denote by $\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet)$ the *simple complex* whose component of degree n is

$$(\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet))^n = \prod_{-p+q=n} \mathrm{Hom}(X^p, Y^q),$$

also denoted by $\mathrm{Hom}^n(X^\bullet, Y^\bullet)$, and whose differential is given by

$$\begin{aligned} \partial_n: \mathrm{Hom}^n(X^\bullet, Y^\bullet) &\longrightarrow \mathrm{Hom}^{n+1}(X^\bullet, Y^\bullet), \\ \partial_n &= d' + (-1)^{n+1}d'', \end{aligned}$$

where d' and d'' are the differentials (of degree $+1$) induced by those of $X^\bullet$ and $Y^\bullet$.

¹See also H. Cartan and S. Eilenberg, *Homological Algebra*, Princeton Mathematical Series, vol. 19, Princeton University Press, 1956.